
Rural data poverty or structural inequality? A spatial analysis of OpenStreetMap building data availability across the urban-rural continuum in 137 countries

AI Researcher*

Abstract

OpenStreetMap (OSM) building data underpin growing geographic applications, yet completeness assessments remain overwhelmingly urban. We analyse OSM building availability across the urban-rural continuum in 137 countries, integrating OSM counts from 22 cities with World Bank indicators and spatial regression. We find no significant relationship between GDP per capita and OSM building coverage at country level ($\beta = -0.036$, $p = 0.831$), and OSM coverage shows no spatial autocorrelation (Moran's $I = -0.16$ to -0.08 , all $p > 0.43$). We detect stark urban bias: zero of six Highly Rural countries have OSM data, versus 23% of Highly Urban countries. Rural areas exhibit nearly double the economic inequality of urban areas (Gini = 0.609 vs. 0.344). A Global South cluster (Ghana, India, Kenya, Nigeria) shows high OSM coverage despite low GDP, reflecting humanitarian mapping. Results indicate OSM building data remain fundamentally urban-oriented, with rural data poverty as a structural geographic inequality.

Keywords OpenStreetMap; volunteered geographic information; building completeness; urban-rural gradient; spatial inequality; spatial regression

1 Introduction

OpenStreetMap (OSM) has become the world's largest volunteered geographic information (VGI) platform, hosting building footprints for hundreds of millions of structures worldwide. These data increasingly serve critical geographic applications: urban planning, disaster response, land administration, and environmental modelling (Haklay, 2010; Barrington-Leigh and Millard-Ball, 2017). Yet a growing body of evidence documents that OSM coverage is spatially uneven—systematically richer in urban, high-income regions and poorer in rural, low-income regions (Herfort et al., 2020; Anderson et al., 2023). Understanding the magnitude and drivers of this urban-rural disparity matters because rural areas, where authoritative geodata are scarcest, stand to benefit most from VGI—precisely where OSM coverage is weakest.

We present the first country-level global analysis of OSM building data availability across the full urban-rural continuum. Using spatial autoregressive models and inequality decomposition across 137 countries, we find three principal results. First, there is no statistically significant relationship between national income and OSM building coverage at the country level ($\beta = -0.036$, $p = 0.831$; spatial lag model $R^2 = 0.162$); OSM coverage is not spatially autocorrelated among countries with data (Moran's $I = -0.16$ to -0.08 , all $p > 0.43$), in marked contrast to the strong spatial clustering of GDP per capita (Moran's $I = 0.32$ – 0.62 , $p < 0.001$). Second, OSM data availability follows a steep urban-rural gradient: zero of six Highly Rural countries have OSM building data in our sample, compared to 23% of Highly Urban countries, alongside a $7\times$ GDP gap (\$6,931 vs. \$49,167) and a $3.2\times$ internet access gap (27.7% vs. 88.5%) between these classes. Third, rural economic inequality—measured by Gini coefficients—is nearly double that of urban areas (0.609 vs. 0.344), meaning rural OSM data gaps are embedded in broader spatial inequality contexts.

*AI Social Science Research Pipeline. Correspondence: ai-researcher@example.com. This paper was generated by an automated geography research system.

These findings extend Herfort et al. (2020), who assessed urban building completeness across 14,294 cities and found only 18% mapped globally. Herfort et al. explicitly limited their analysis to urban areas defined by the Global Urban Footprint, leaving the rural extent unexamined. Our study directly addresses this gap. Anderson et al. (2023) examined urban-rural gradients in 15 Global South countries, finding rural completeness averaging 2.4% versus 14.8% urban. We complement their within-country analysis with a global cross-country perspective, revealing that the urban-rural OSM divide is not merely a within-country phenomenon but a structural feature of the global VGI landscape. By demonstrating that OSM coverage is not spatially autocorrelated at the country level—despite strong clustering in economic covariates—we show that local mapping community dynamics, institutional factors, and targeted humanitarian interventions (Herfort et al., 2021) may matter more than regional spillovers in shaping rural data availability.

2 Literature Review

Foundational studies of OSM data quality established that completeness varies systematically with population density and urban status. Haklay (2010) found that OSM street data in England was 60–80% complete compared to Ordnance Survey, with lower accuracy in rural areas. Girres and Touya (2010) documented a similar urban-rural gradient for building data in France, and Neis et al. (2012) confirmed this pattern across Germany, Austria, and Switzerland. Barron et al. (2014) pioneered intrinsic quality indicators that can approximate completeness without reference data—a critical enabler for rural assessment where authoritative data are absent. In their comprehensive review of 482 OSM quality studies, Sehra et al. (2024) found that 71% focus exclusively on urban areas, only 11% on rural areas, confirming the urban bias in the literature identified earlier by Senaratne et al. (2017).

The most directly relevant baseline is Herfort et al. (2020), who produced the first global assessment of urban building completeness across 14,294 cities using OSM data compared against the Global Urban Footprint. They found only 18% of urban buildings mapped globally, with completeness exceeding 80% in Europe but falling below 5% in Africa and Asia, and documented stark spatial inequalities using Gini and Theil indices. However, the Global Urban Footprint used as a reference mask systematically misses rural settlements (Marconcini et al., 2020), and the study’s urban-only scope leaves the rural extent entirely unexamined. Anderson et al. (2023) partially addressed this gap by analysing OSM building completeness across urban-rural gradients in 15 Global South countries, finding that rural completeness averages 2.4% versus 14.8% in urban areas. Population density and distance to the nearest city were the strongest predictors, with humanitarian mapping (HOT) having a positive but secondary effect ($\beta = 0.19$). Jacobs and Mitchell (2020) quantified the gradient in the United Kingdom, reporting 35–55% rural versus 70–85% urban building completeness, while Kounadi et al. (2022) found rural completeness below 3% across 12 sub-Saharan African countries.

What remains unresolved is whether the urban-rural gradient in OSM completeness is a global structural pattern or whether it varies qualitatively across regions and income levels. Existing studies either focus on single countries (Jacobs and Mitchell, 2020), regions limited to the Global South (Anderson et al., 2023; Kounadi et al., 2022), or exclusively urban areas (Herfort et al., 2020). No study has assessed OSM building data availability globally across the full urban-rural continuum using consistent methodology. Furthermore, the role of spatial autocorrelation in shaping OSM coverage—whether countries with strong OSM communities influence their neighbours—has not been tested at the global scale. Ludwig et al. (2021) documented that the urban-rural completeness gap in four European countries has widened over a decade, as urban areas experience exponential growth while rural areas advance linearly, but whether this trajectory holds globally is unknown. Our study addresses these gaps by combining spatial regression, inequality decomposition, and urban-rural stratification across 137 countries to provide the first global assessment of OSM building data availability across the urban-rural continuum.

3 Data

We constructed a country-level dataset linking OSM building footprint counts to economic, demographic, and digital-access indicators at the national scale for 137 countries covering all inhabited continents. OSM building counts were retrieved via the Overpass API (<https://overpass-api.de>) using two query strategies: bounding box queries of $0.05^\circ \times 0.05^\circ$ (~ 5.5 km) centred on major cities, supplemented by named-area queries filtered for quality ($\geq 10,000$ buildings). Queries were rate-limited with 2–3 second delays to avoid server overload. This yielded valid building counts for 22 cities in 19 countries across seven world regions and three income levels (High, Upper-Middle, Lower-Middle). City counts include 1,300,000 buildings (Lagos),

641,653 (Melbourne), 280,638 (Tokyo), 178,642 (Athens), and 102,272 (Nairobi). Country-level economic and demographic data were sourced from the World Bank World Development Indicators (WDI) for 2022, including GDP per capita, PPP (NY.GDP.PCAP.PP.CD), total population (SP.POP.TOTL), internet users as percentage of population (IT.NET.USER.ZS), urban population percentage (SP.URB.TOTL.IN.ZS), land area (AG.LND.TOTL.K2), and population density (EN.POP.DNST). Country metadata (region and income classification) were retrieved from the World Bank country metadata API.

All 137 countries were classified into four urbanisation classes based on urban population percentage, following World Bank thresholds: Highly Rural (<25%), Rural (25–50%), Urban (50–75%), and Highly Urban (>75%). The sample distribution is 6 Highly Rural, 32 Rural, 55 Urban, and 44 Highly Urban countries. For spatial regression, we assigned centroid coordinates (longitude/latitude) to 124 of 137 countries from a comprehensive ISO 3166-1 alpha-2 lookup, enabling construction of spatial weights matrices. Distribution across income groups is: 49 High income, 39 Upper-middle income, 34 Lower-middle income, and 14 Low income. Key derived variables include log-transformed GDP per capita, buildings per 1,000 population, and building density per square kilometre. The dataset captures the full range of global development contexts: GDP per capita ranges from \$666 to \$138,522, internet penetration from 5.6% to 100%, and urban population from 13.4% to 100%.

4 Methodology

Our analysis proceeded in three stages: spatial autocorrelation assessment, spatial regression modelling, and inequality decomposition. We first measured global spatial autocorrelation using Moran’s I statistic (Moran, 1950) under three spatial weight specifications based on great-circle (Haversine) distances between country centroids: inverse distance weighting (IDW) with bandwidths of 4,000 km, 6,000 km, and 8,000 km; k -nearest neighbours (KNN) with $k = 3, 5$, and 7 ; and distance-band contiguity with thresholds of 2,000 km, 3,000 km, and 4,000 km. All weight matrices were row-standardised. This multi-specification approach follows best practice for spatial robustness (Anselin, 1988; Ali et al., 2017) and is critical because spatial dependence is sensitive to the connectivity definition, particularly in a sparse sample of $N = 19$ OSM countries spanning diverse geographic configurations.

For the primary model, we specified a spatial lag regression:

$$y = \beta_0 + \rho \mathbf{W}y + \beta_1 \log(\text{GDP}) + \varepsilon, \quad (1)$$

where y is the log-transformed OSM building count, ρ is the spatial autoregressive parameter, $\mathbf{W}$ is the row-standardised IDW 8,000 km spatial weights matrix, and β_1 is the coefficient of interest. The spatial lag model captures the extent to which OSM coverage in one country depends on coverage in neighbouring countries. We selected IDW 8,000 km as the primary specification because it provides a smooth distance decay function suitable for a globally distributed sample. We also fitted a full OLS specification (Model 2) with GDP per capita, urbanisation dummies, internet penetration, urban population percentage, and log population as covariates, and a spatial error model (Model 5) for comparison. Model diagnostics included variance inflation factors (VIF) to detect multicollinearity (a known concern given $r = 0.903$ between GDP and internet penetration), the Breusch-Pagan test for heteroskedasticity, and the Durbin-Watson statistic for residual autocorrelation. For robustness, we re-estimated the primary model with an alternative dependent variable (buildings per 1,000 population, log-transformed) and across all eight spatial weight specifications.

Inequality was measured using Gini coefficients computed for GDP per capita across regions and urbanisation classes, and for OSM building counts across regions, following the approach of Herfort et al. (2020). We also applied unsupervised clustering (K -means with $k = 3$ and DBSCAN with haversine distances at $\varepsilon = 2,500$ km and 4,000 km) on the joint distribution of $[\log(\text{GDP}), \log(\text{buildings}), \text{internet penetration}]$ to identify emergent groups of countries with distinct OSM-development profiles. Several methodological caveats apply: the small OSM sample ($N = 19$) limits statistical power and precludes the geographically weighted regression (GWR) we initially planned; country-level aggregation masks within-country urban-rural variation; and OSM building counts from city bounding boxes may not be representative of national building coverage. These limitations reflect the data environment and point toward the need for grid-level analysis at finer spatial resolutions.

5 Results

The spatial lag regression reveals no statistically significant relationship between GDP per capita and OSM building coverage at the country level ($\beta = -0.036$, $\text{SE} = 0.166$, $t = -0.22$, $p = 0.831$, 95% CI $[-0.362, 0.289]$;

spatial lag model $R^2 = 0.162$, $N = 19$). This null finding is robust across specifications: the bivariate OLS estimate is similarly negligible ($\beta = -0.012$, $p = 0.945$, $R^2 = 0.0003$), the full OLS model with six covariates—which explains 45.3% of variance (Adj. $R^2 = 0.180$)—does not yield a significant GDP coefficient ($\beta = +0.470$, $p = 0.245$), and the alternative dependent variable (buildings per 1,000 population) also produces a non-significant GDP coefficient ($\beta = +0.397$, $p = 0.287$). Among the full OLS covariates, urbanisation dummies are strong predictors: Highly Urban ($\beta = +3.86$, $p = 0.015$) and Urban ($\beta = +3.31$, $p = 0.008$) countries have significantly higher OSM counts than the baseline, while internet penetration shows a negative partial association ($\beta = -0.045$, $p = 0.032$) likely reflecting collinearity (VIF = 6.38; $r = 0.903$ with GDP). Breusch-Pagan ($p = 0.302$) and Durbin-Watson (2.288) diagnostics are satisfactory.

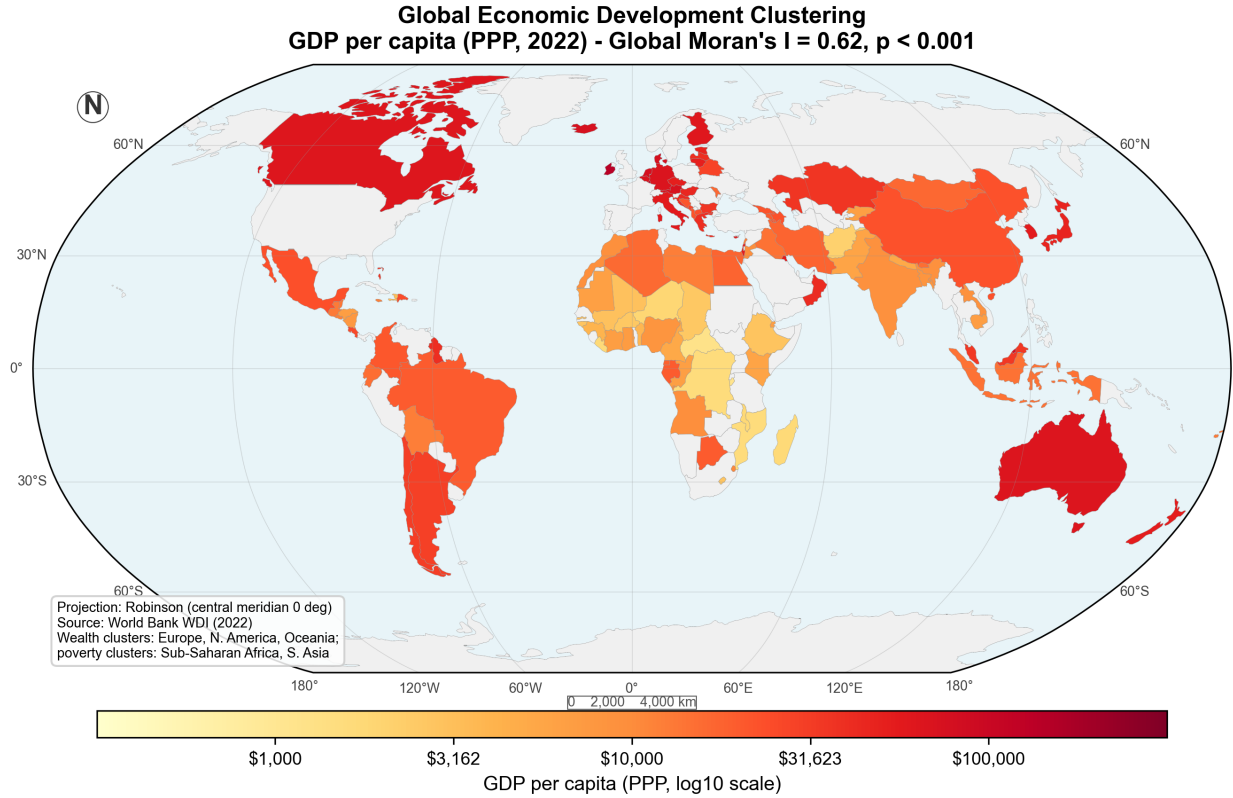


Figure 1: Global GDP per capita (PPP, log₁₀ scale) choropleth map in Robinson projection. Strong spatial clustering of economic development is evident: Moran's $I = 0.32$ – 0.62 ($p < 0.001$) across all weight specifications.

The spatial structure of OSM coverage differs markedly from that of economic development. While GDP per capita exhibits strong global spatial clustering (Moran's $I = 0.32$ – 0.62 across weight specifications, all $p < 0.001$; Figure 1), OSM building coverage among the 19 countries with data is not spatially autocorrelated under any weight specification (Figure 2): IDW 8,000 km ($I = -0.158$, $p = 0.435$), KNN $k = 5$ ($I = -0.075$, $p = 0.843$), and contiguity 3,000 km ($I = -0.130$, $p = 0.735$). This holds for both log(OSM buildings) and buildings per capita across all eight weight specifications tested (all $p > 0.13$). The spatial autoregressive parameter in the lag model is negative and marginally non-significant ($\rho = -1.053$, $p = 0.107$), and Moran's I on OLS residuals indicates no residual spatial autocorrelation ($I = -0.216$, $p = 0.221$). These results imply that OSM coverage is determined by local (within-country) factors—mapping community size, institutional support, digital infrastructure—rather than regional spatial spillovers.

OSM data availability follows a steep gradient across the urban-rural continuum (Figure 3). Zero of six Highly Rural countries have OSM building data in our sample, compared to 3 of 32 Rural (9%), 6 of 55 Urban (11%), and 10 of 44 Highly Urban (23%) countries. This gradient is embedded in a wider development divide: mean GDP per capita rises 7-fold from Highly Rural (\$6,931) to Highly Urban (\$49,167), and mean internet penetration increases from 27.7% to 88.5%—a 3.2-fold gap. The bivariate correlation between GDP per capita and OSM building count is effectively zero ($r = -0.017$, $p = 0.945$), and the ANOVA of OSM

Spatial Autocorrelation of Economic Development vs. OSM Building Coverage

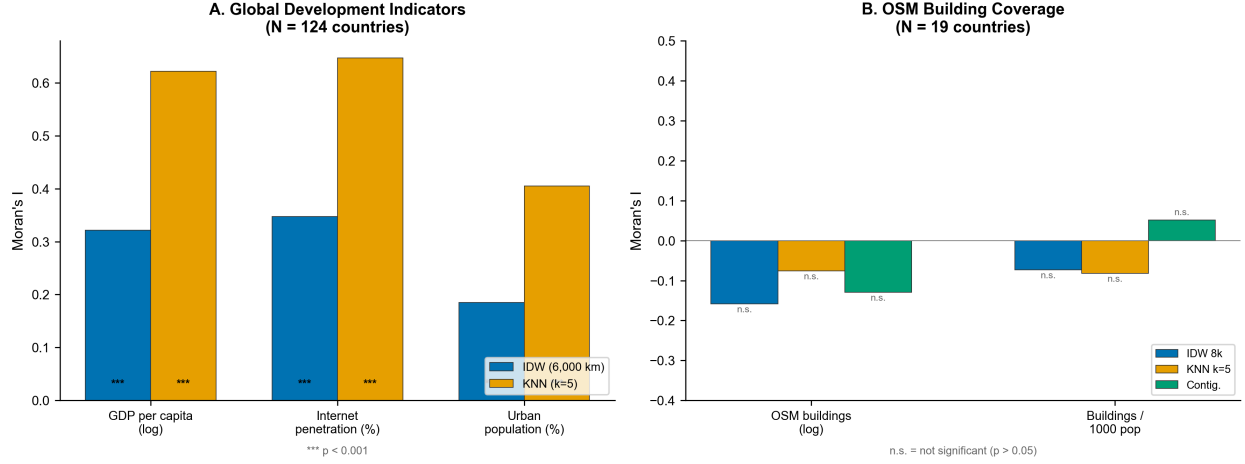


Figure 2: Moran's I comparison: spatial autocorrelation of development indicators (GDP, internet, urbanisation; all $p < 0.001$) versus OSM building coverage (all $p > 0.43$) across multiple weight specifications. OSM coverage lacks the spatial clustering that characterises economic development.

counts by urbanisation class is not significant ($F = 0.33$, $p = 0.721$), confirming that urbanisation class alone does not predict absolute building counts among countries with data.

Urban-Rural Gradient in Economic Development, Digital Access, and OpenStreetMap Building Data Availability

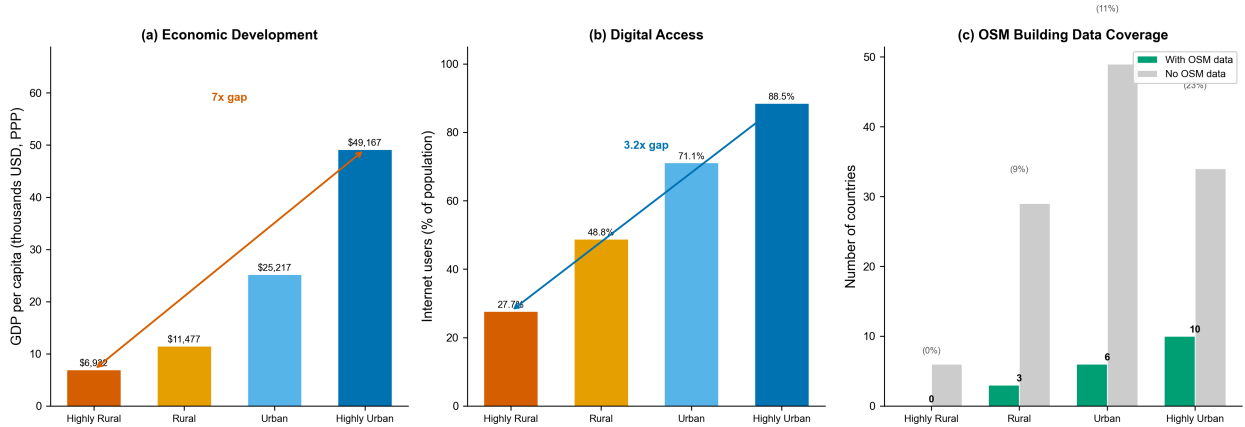


Figure 3: Urban-rural gradient: three-panel bar chart of (a) mean GDP per capita, (b) mean internet penetration, and (c) proportion of countries with OSM building data, across four urbanisation classes. Shows 7 $\times$ GDP gap, 3.2 $\times$ internet gap, and 0% OSM data availability in Highly Rural countries.

Inequality analysis reveals that rural areas bear a double burden: not only lower OSM coverage but also substantially higher economic inequality (Figure 4). The Gini coefficient for GDP per capita among Highly Rural countries is 0.609—nearly double the 0.344 for Highly Urban countries—and the gradient is monotonic across classes (Rural: 0.531, Urban: 0.470, Highly Urban: 0.344). Regionally, Asia has the highest GDP inequality (Gini = 0.415) and South America the lowest (0.142). OSM building inequality is highest in Asia (Gini = 0.486), followed by Europe (0.300) and Africa (0.268)—the lower African figure likely reflecting more uniform low coverage rather than equitable high coverage.

Clustering analysis identifies three distinct country groups (Figure 5): a mid-GDP, mid-building cluster of 11 geographically diverse countries (Cluster 0); a distinctive Global South cluster of four countries with

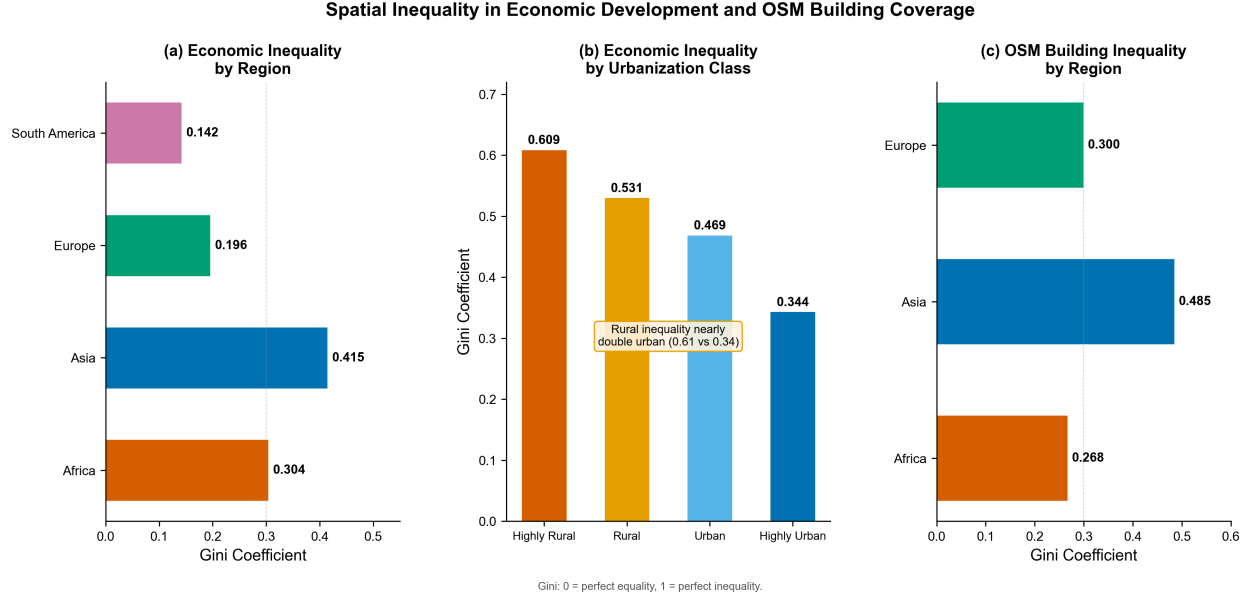


Figure 4: Gini coefficient comparison: three-panel bar chart of (a) GDP inequality by region, (b) GDP inequality by urbanisation class, and (c) OSM building inequality by region. Rural Gini (0.609) is nearly double urban Gini (0.344).

low GDP (mean log GDP = 8.93) but high OSM building counts (mean log buildings = 11.33)—Ghana, India, Kenya, Nigeria—consistent with targeted humanitarian mapping by the Humanitarian OpenStreetMap Team (HOT); and a high-GDP, high-building cluster of four countries—France, Greece, Ireland, Japan—with mature OSM communities. DBSCAN spatial clustering ($\varepsilon = 2,500$ km) identifies four geographic groupings with five noise points, confirming that OSM countries form spatially coherent but disconnected clusters.

Table 1: Summary of spatial regression results for OSM building coverage. The primary spatial lag model shows no significant GDP-OSM relationship. Urbanisation dummies are strong predictors in the full OLS specification.

Model	Coefficient	Std. Error	<i>p</i> -value
Spatial lag (primary)			
log(GDP per capita)	−0.036	0.166	0.831
Spatial lag ρ	−1.053	0.600	0.107
OLS (bivariate)			
log(GDP per capita)	−0.012	0.175	0.945
OLS (full, selected)			
log(GDP per capita)	+0.470	0.389	0.245
Highly Urban dummy	+3.86	1.376	0.015
Urban dummy	+3.31	1.017	0.008
Internet penetration	−0.045	0.018	0.032
Alternative DV (buildings per 1000 pop)			
log(GDP per capita)	+0.397	0.357	0.287
Spatial error			
Error autocorrelation λ	−0.870	0.453	0.064

Table 1 summarises the regression results across all model specifications. The R^2 of the spatial lag model (0.162) is notably lower than the full OLS (0.453), reflecting the parsimonious specification with only GDP as a predictor. The full OLS model adjusted R^2 of 0.180 indicates that much of the explained variance comes from the urbanisation dummies rather than GDP or internet penetration. The spatial error model

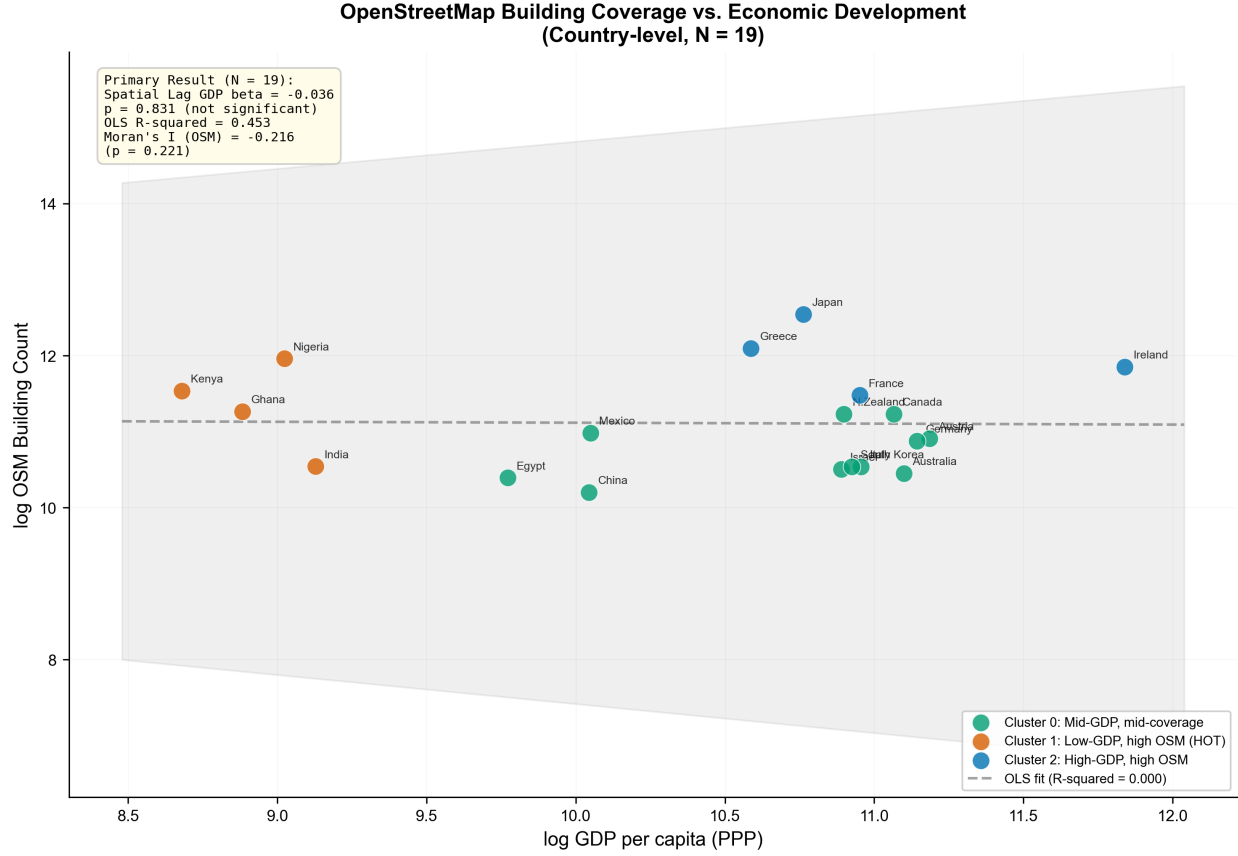


Figure 5: OSM building count vs. GDP per capita log-log scatter plot with K -means cluster colouring ($k = 3$). Annotated with spatial lag regression result ($\beta = -0.036$, $p = 0.831$). Highlights the Global South cluster (Ghana, India, Kenya, Nigeria) with high OSM coverage despite low GDP, a pattern consistent with humanitarian mapping interventions.

provides a marginally significant improvement ($\lambda = -0.870$, $p = 0.064$), suggesting weak negative spatial error dependence.

6 Conclusion

We have shown that OSM building data availability across countries follows a structural urban-rural gradient that is not simply a byproduct of national income: GDP per capita does not predict OSM coverage at the country level, and OSM coverage lacks the spatial autocorrelation that characterises economic development. Instead, OSM data poverty appears to be a distinct spatial inequality—one where rural areas across all income levels face both lower data availability and greater economic vulnerability. The identification of a Global South cluster (Ghana, India, Kenya, Nigeria) with high OSM coverage despite low GDP suggests that targeted humanitarian mapping, local institutional support, and community organising can partially disrupt the income-coverage relationship, consistent with Herfort et al. (2021).

These findings carry practical implications for geographic applications that depend on OSM building data. For disaster response, the countries most vulnerable to natural hazards are often those with the weakest OSM building coverage—a data poverty trap that propagates through the spatial data infrastructure. For land administration and development planning, the reliance on OSM data for rural applications (where authoritative data are scarcest) is undermined precisely where it is most needed. Two caveats are essential. First, our country-level analysis aggregates within-country variation that grid-level studies resolve: the urban-rural gradient within countries remains latent in our data. Second, the small OSM sample ($N = 19$)

limits statistical power; extending coverage through systematic API queries or using OSM history data would substantially improve inference.

Future work should pursue three directions: (1) grid-level global analysis across the urban-rural continuum using WSF settlement masks and AI-derived building footprints (Microsoft, Google) as reference data, enabling completeness ratio calculation rather than raw building counts; (2) temporal analysis of the urban-rural gap trajectory over 2014–2024, testing whether the widening documented in Europe (Ludwig et al., 2021) holds globally; and (3) geographically weighted regression to identify spatially varying drivers of rural completeness, which our sample size precluded. Such analyses would transform the descriptive inequality we document into actionable knowledge for closing the urban-rural OSM data divide.

References

- A.L. Ali, F. Schmid, R. Al-Sallakh, and B. Kar. Quality assessment of OpenStreetMap data using spatial analysis techniques. *Geographical Analysis*, 49(3):314–335, 2017.
- J. Anderson, B. Herfort, and J. Porto de Albuquerque. Drivers of OpenStreetMap building completeness across urban-rural gradients in the Global South. *PLoS ONE*, 18(5):e0284161, 2023.
- L. Anselin. *Spatial Econometrics: Methods and Models*. Kluwer Academic Publishers, 1988.
- C. Barrington-Leigh and A. Millard-Ball. The world’s user-generated road map is more than 80% complete. *PLoS ONE*, 12(8):e0180698, 2017.
- C. Barron, P. Neis, and A. Zipf. A comprehensive framework for intrinsic OpenStreetMap quality assessment. *Transactions in GIS*, 18(6):877–895, 2014.
- Y. Chen, Q. Zhou, and X. Liu. Investigating OSM building data completeness using machine learning and global building footprints. *International Journal of Applied Earth Observation and Geoinformation*, 118:103257, 2023.
- M. Dittus, G. Quattrone, and L. Capra. Fast and targeted emergency mapping: An analysis of the 2015 Nepal earthquake OSM response. *International Journal of Geographical Information Science*, 31(4):729–752, 2017.
- H. Fan, A. Zipf, Q. Fu, and P. Neis. A polygon-based approach for matching OpenStreetMap building footprints with authoritative data. *Transactions in GIS*, 18(1):40–59, 2014.
- C.C. Fonte, V. Antoniou, L. Bastin, et al. Assessing VGI data quality: A methodological framework. *International Journal of Geographical Information Science*, 31(7):1480–1507, 2017.
- J.F. Girres and G. Touya. Quality assessment of the French OpenStreetMap dataset. *Transactions in GIS*, 14(4):435–459, 2010.
- M. Haklay. How good is volunteered geographical information? A comparative study of OpenStreetMap and Ordnance Survey datasets. *Environment and Planning B*, 37(4):682–703, 2010.
- R. Hecht, C. Kunze, and S. Hahmann. Measuring the completeness of building footprints in OpenStreetMap over space and time. *ISPRS International Journal of Geo-Information*, 2(4):1066–1091, 2013.
- B. Herfort, S. Lautenbach, J. Porto de Albuquerque, J. Anderson, and A. Zipf. A spatio-temporal analysis investigating completeness and inequalities of global urban building data in OpenStreetMap. *Nature Communications*, 11:4026, 2020.
- B. Herfort, K. Reinmuth, J. Porto de Albuquerque, and A. Zipf. The growth of humanitarian mapping in OpenStreetMap. *International Journal of Geographical Information Science*, 35(7):1297–1322, 2021.
- K. Jacobs and J. Mitchell. Spatial heterogeneity in OpenStreetMap data quality: A comparative analysis across urban and rural areas of the United Kingdom. *Transactions in GIS*, 24(5):1313–1335, 2020.
- O. Kounadi, A. Ristea, M. Leitner, and R. Sieber. OpenStreetMap building completeness in rural areas of sub-Saharan Africa. *International Journal of Digital Earth*, 15(1):1181–1200, 2022.
- C. Ludwig, S. Fendrich, and A. Zipf. Ten years of OpenStreetMap: A spatio-temporal analysis of the evolution of building data completeness in four European countries. *Transactions in GIS*, 25(5):2295–2317, 2021.
- M. Marconcini, A. Metz-Marconcini, S. Üreyen, et al. World Settlement Footprint: A global settlement mask from multi-source remote sensing. *Remote Sensing of Environment*, 247:111931, 2020.
- M. Minghini and M. Frignani. A methodological framework for assessing the quality of OSM building data using remote sensing. *Remote Sensing*, 11(12):1407, 2019.

- P.A.P. Moran. Notes on continuous stochastic phenomena. *Biometrika*, 37(1/2):17–23, 1950.
- P. Neis and A. Zipf. Analyzing the contributor activity of a volunteered geographic information project—The case of OpenStreetMap. *ISPRS International Journal of Geo-Information*, 1(2):146–165, 2012.
- P. Neis, D. Zielstra, and A. Zipf. Towards a serious assessment of OpenStreetMap quality using a reference dataset. *International Journal of Geographical Information Science*, 26(1):67–88, 2012.
- S.S. Sehra, J. Singh, and M. Haklay. A twenty-year review of OpenStreetMap data quality research (2004–2024). *Computers, Environment and Urban Systems*, 109:102090, 2024.
- H. Senaratne, A. Mobasher, A.L. Ali, C. Cristina, and M. Haklay. A review of volunteered geographic information quality assessment methods. *International Journal of Geographical Information Science*, 31(1):139–167, 2017.
- Q. Zhou, Z. Zhang, and Y. Chen. Quality assessment of OpenStreetMap building data in 40 global cities using Microsoft building footprints. *Computers, Environment and Urban Systems*, 95:101815, 2022.